CALIBRATION OBJECTIVES FOR TRANSFERRED SPECULATIVE DRAFTERS

PRELIMINARY EVIDENCE REPORT

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ABSTRACT

We investigate whether acceptance-aware token supervision improves a transferred feature-conditioned speculative drafter. This working report records completed development experiments from an ongoing study, rather than a finished submission. With 4,096 training records and three epochs, the reproduced feature objective outperformed the token-loss configurations in the earlier development comparison. A separate 512-anchor study crosses adapter architecture with CE and AUF; its final results are not yet available. Additional drafter LoRA training from the feature-fitted checkpoint gives near-parity throughput in its first full timing. Repeated timing, broader workloads, scaling, and confirmation experiments remain unfinished.

1 QUESTION AND CONTROLLED INTERVENTIONS

RelaySpec transfers a frozen feature-conditioned drafter to a new target through a learned linear interface. We compare the reproduced ZIP feature objective with ordinary token cross-entropy (CE) and accept-until-fail (AUF) cross-entropy while keeping the original targets frozen. No target-side LoRA adaptation is used. Equal record counts and epochs do not imply equal optimizer updates, supervised positions, or compute across feature and token objectives.

For valid supervised positions $m_{b,j}$, draft probabilities $p_{\theta,b,j}$, and labels $y_{b,j}$ from the original target's rollout, the detached AUF support is

$$w_{b,j} = \mathrm{stopgrad}\left[\prod_{i<j}\Big(1 - m_{b,i} + m_{b,i}\mathbf{1}[\arg\max_{v} p_{\theta,b,i}(v) = y_{b,i}]\Big)\right], \tag{1}$$

$$\mathcal{L}_{\mathrm{AUF}} = -\frac{\sum_{b,j} m_{b,j} w_{b,j} \log p_{\theta,b,j}(y_{b,j})}{\sum_{b,j} m_{b,j} w_{b,j}}. \tag{2}$$

The clean anchor and padding are excluded. The first incorrect proposal remains supervised; positions after it do not. Normalization is performed inside each microbatch, followed by equal microbatch averaging per update. Pure AUF is used for this token intervention; no simultaneous MSE or KL term is added. ZIP denotes the reproduced layer and normalized fused-context feature objective, with its original position sampling and example weighting.

2 EARLIER DEVELOPMENT EVIDENCE

Each result below uses 128 Numina development requests, a maximum of 2,048 generated tokens with natural EOS, fitting seed42, and one warmed timing. The inference backend is vLLM0.28.0 on L40S hardware. These are development measurements, not untouched confirmation results. Exactness means equality of the complete generated token sequence with the paired autoregressive (AR) output. It does not establish mathematical correctness of an answer or a universal guarantee across numerical configurations.

Target	Objective	Tokens/s	Speedup	Exact/128
Qwen3-8B	AR	24.56	1.00	reference
	ZIP	173.94	7.08	128
	CE	108.75	4.43	128
	AUF	98.81	4.02	128
Llama-3.2-3B	AR	46.01	1.00	reference
	ZIP	186.93	4.06	128
	CE	142.23	3.09	128
	AUF	112.86	2.45	128

Table 1: Completed loss replacements. Qwen3-4B and Llama-3.1-8B drafters serve the listed targets. Speedup is relative to each target's own AR run. Differences in target and runtime configuration preclude interpreting cross-family absolute throughput as a model ranking.

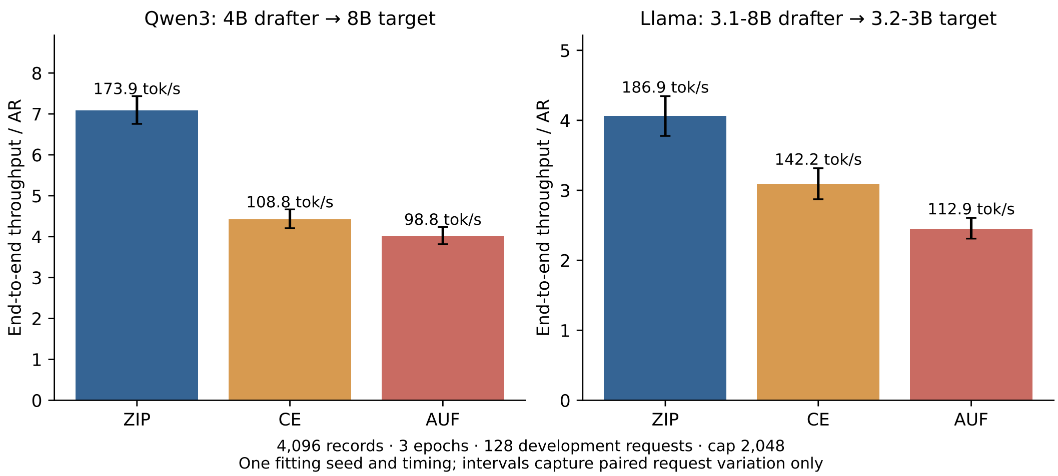

Figure 1: Throughput after fitting each objective. Intervals use 10,000 paired request bootstrap samples; they exclude fitting-seed, timing-order and selection uncertainty. The current AUF intervention reduces throughput relative to feature fitting in both completed settings.

3 Does adapting the drafter add value?

We additionally hold the trained ZIP interface fixed and adapt drafter attention and MLP projections with rank32 LoRA, using either CE or AUF. All branches reuse the same 4,096 records. The initial ZIP fit receives three epochs and each continuation receives three additional epochs. This is an incremental adaptation test; it is not an equal-total-compute comparison with the unchanged ZIP checkpoint. No source transformer is restored during decoding.

Qwen adaptation	Tokens/s	Ratio to ZIP	Exact/128
ZIP (earlier timing)	173.94	1.000	128
ZIP + drafter LoRA CE	173.10	0.995	128
ZIP + drafter LoRA AUF	174.83	1.005	128

Table 2: First complete drafter-LoRA timing, compared with the earlier ZIP timing. Three fresh unchanged ZIP/native controls and repeated LoRA timings are pending. These approximately half-percent differences do not establish a winner.

Merged-checkpoint logits agree exactly with the training implementation. The vLLM attachment check also verifies full projection tensors and the derived fused context-KV buffer against the exported weights. All reported LoRA outputs match paired AR outputs on 128/128 requests.

4 Scope and unfinished evidence

Our reported Qwen3 experiments use a shared tokenizer and token-ID vocabulary; the mapper adapts hidden representations or the fusion interface, not vocabulary IDs. Consequently, these results do not demonstrate heterogeneous-vocabulary DFlash support.

Qwen14 evaluation, multiple-workload measurements, timing and fitting-seed replication, data and compute scaling, capacity, regularization, untouched confirmation, and the final Transformers replications remain incomplete. Related work, contribution claims and the full discussion will be written against the completed evidence. This report does not claim a new state-of-the-art method or submission readiness. Machine-readable claim and source hashes are recorded in `claim_evidence.json`.

5 Controlled 512-anchor study in progress

The new study separates initialization, architecture, and token objective. Original RelaySpec and ZIP are feature-fitting procedures, not CE losses. Their unchanged checkpoints remain controls for CE and AUF continuations. Interfaces include the normalized RelaySpec map, a residual fusion adapter, five dense layer maps, one unrestricted fusion matrix, and five independent rank-56 residual layer adapters. The source drafter and target remain frozen.

DFlash's Appendix A.1 reports 512 sampled anchors per sequence and position-decayed CE[1]. Uniform CE and AUF are explicit objective interventions. Our protocol uses 4,096 cached records, a 4,096-token rollout cap, and 16,000 presentations at global batch eight. Short records supply fewer distinct anchors. Tuning uses a separate 32-request validation set; final comparisons require 128 requests, a 2,048-token cap, and three timing repetitions.

The first Qwen8 five-BA fits completed 100 updates (800 presentations) with 1,863,680 trainable parameters. AUF took 337.8 seconds and CE 339.7 seconds on two L40S GPUs; both peaked near 17.04 GB allocated memory on rank zero. These incremental fitting costs exclude ZIP initialization and rollouts. Both exports passed frozen-weight and folding checks. Their decoding validation is pending: completed four-request exactness checks used separate two-update pilots, not these 100-update checkpoints. No decoding advantage for either loss is established by these costs.

Qwen-to-Llama additionally requires source-token labels, causal target-feature alignment, and proposal retokenization. Source-token AUF support is not target-token accepted progress. Cross-family results require validation of the complete decoding path.

[1] `https://arxiv.org/html/2602.06036v1`